

Delegates

Create a class to perform the mathematical operations such as Add, Multiply, Subtract, Divide. Create a delegate to call these methods from the main()

```
using System;
```

```
delegate double MathOperation(double a, double b);
```

```
class MathOperations
```

```
{
```

```
    public double Add(double a, double b)
```

```
    {
```

```
        return a + b;
```

```
    }
```

```
    public double Subtract(double a, double b)
```

```
    {
```

```
        return a - b;
```

```
    }
```

```
    public double Multiply(double a, double b)
```

```
    {
```

```
        return a * b;
```

```
    }
```

```
    public double Divide(double a, double b)
```

```
    {
```

```
        if (b != 0)
```

```
            return a / b;
```

```
        else
```

```
        {
```

```
            Console.WriteLine("Division by zero is not allowed.");
```

```
            return 0;
```

```

    }
}
}
class Program
{
    static void Main()
    {
        MathOperations obj = new MathOperations();
        MathOperation operation;
        Console.Write("Enter First Number: ");
        double num1 = Convert.ToDouble(Console.ReadLine());
        Console.Write("Enter Second Number: ");
        double num2 = Convert.ToDouble(Console.ReadLine());
        Console.WriteLine("\n1. Add");
        Console.WriteLine("2. Subtract");
        Console.WriteLine("3. Multiply");
        Console.WriteLine("4. Divide");
        Console.Write("Enter your choice: ");
        int choice = Convert.ToInt32(Console.ReadLine());
        switch (choice)
        {
            case 1:
                operation = new MathOperation(obj.Add);
                Console.WriteLine("Addition = " + operation(num1, num2));
                break;
            case 2:
                operation = new MathOperation(obj.Subtract);
                Console.WriteLine("Subtraction = " + operation(num1, num2));

```

```

        break;
    case 3:
        operation = new MathOperation(obj.Multiply);
        Console.WriteLine("Multiplication = " + operation(num1, num2));
        break;
    case 4:
        operation = new MathOperation(obj.Divide);
        Console.WriteLine("Division = " + operation(num1, num2));
        break;
    default:
        Console.WriteLine("Invalid Choice");
        break;
    }
    Console.ReadLine();
}
}

```

Output:

```

PS C:\Users\K.Poojitha\OneDrive\Desktop\wipro\Datatypes> dotnet run
Enter First Number: 38900
Enter Second Number: 468788

CALCULATOR MENU
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 3
Result = 18235853200

```